

Use Case Diagram Report

E-Commerce Web Application

Course: Software Engineering Lab

Version: 1.0

Group Members:

- Iftekhar Sourav — 23524202107
- Musfiqur Rahman Sama — 23524202117
- Towfiq Omar Rakin — 23524202131
- Zarin Tasnim Rahman Tuly — 23524202139

1. Introduction

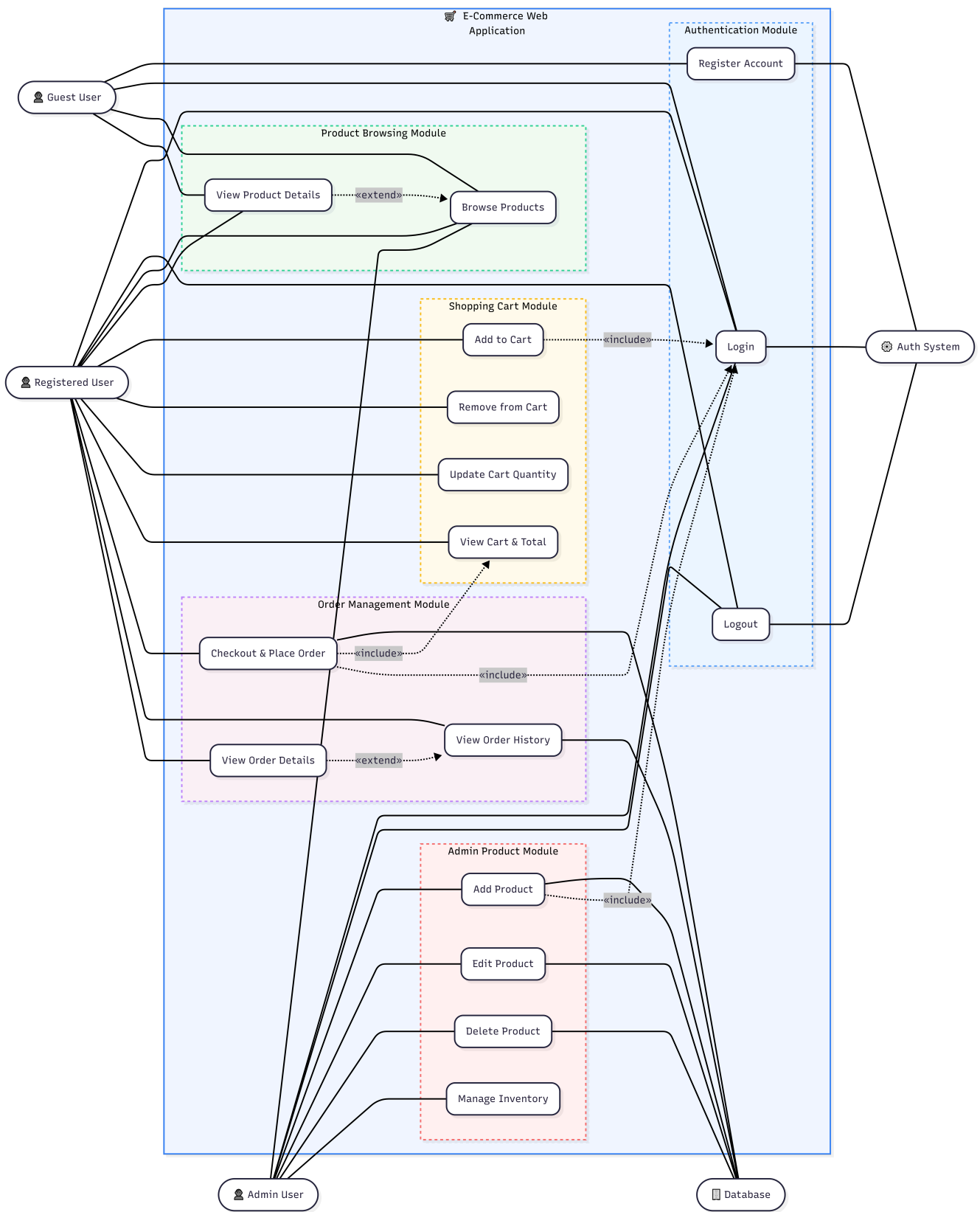
A **Use Case Diagram** is a behavioral UML diagram that captures the functional requirements of a system by illustrating the interactions between external actors and the system's use cases. Each use case represents a goal that an actor can achieve through the system.

This report presents the use case diagram for the **E-Commerce Web Application**, derived directly from the functional requirements documented in the SRS v1.0.

2. Actors

Actor	Type	Description
Guest User	Primary	Unauthenticated visitor who can browse but not purchase
Registered User	Primary	Authenticated customer with full shopping capabilities
Admin User	Primary	Privileged user responsible for product and inventory management
Authentication System	Secondary	Django's built-in auth system handling sessions and credentials
Database	Secondary	Relational data store (SQLite/PostgreSQL) persisting all records

3. Use Case Diagram



4. Use Case Descriptions

UC1 — Register Account

Field	Detail
Actor	Guest User

Field	Detail
Precondition	User is not logged in
Main Flow	User fills in username, email, and password → System validates uniqueness → Password is hashed → Account created
Alternate Flow	Username already exists → System shows error
Postcondition	New user account persisted in the database
FR References	FR1, FR2, FR3, FR4

UC2 — Login

Field	Detail
Actor	Guest User, Registered User, Admin User
Precondition	User has a registered account
Main Flow	User enters credentials → Auth System validates → Session created
Alternate Flow	Invalid credentials → Error message displayed
Postcondition	Authenticated session maintained
FR References	FR5, FR6, FR7

UC3 — Logout

Field	Detail
Actor	Registered User, Admin User
Precondition	User is logged in
Main Flow	User clicks logout → Session destroyed
Postcondition	User returned to guest state
FR References	FR8

UC4 — Browse Products

Field	Detail
Actor	Guest User, Registered User, Admin User
Precondition	None
Main Flow	System displays paginated product listing with names, prices, and availability
Postcondition	User views available products

Field	Detail
FR References	FR13, FR15, FR16

UC5 — View Product Details (*extends UC4*)

Field	Detail
Actor	Guest User, Registered User
Precondition	User is on product listing page
Main Flow	User clicks a product → System displays name, description, price, image, and stock
Postcondition	Detailed product information shown
FR References	FR14, FR15

UC6 — Add to Cart (*includes UC2*)

Field	Detail
Actor	Registered User
Precondition	User is logged in; product is in stock
Main Flow	User clicks "Add to Cart" → Item appended to session cart
Alternate Flow	User not logged in → Redirected to login
Postcondition	CartItem record created/updated
FR References	FR17, FR21

UC7 — Remove from Cart

Field	Detail
Actor	Registered User
Precondition	Cart contains at least one item
Main Flow	User clicks remove → CartItem deleted → Total recalculated
Postcondition	Item removed; total updated
FR References	FR18, FR20

UC8 — Update Cart Quantity

Field	Detail
Actor	Registered User
Precondition	Item exists in cart
Main Flow	User adjusts quantity → System updates CartItem and recalculates total
Postcondition	Cart reflects updated quantity
FR References	FR19, FR20

UC9 — View Cart & Total

Field	Detail
Actor	Registered User
Precondition	User is logged in
Main Flow	User navigates to cart → System displays all items with dynamic total
Postcondition	Cart summary visible to user
FR References	FR20, FR21

UC10 — Checkout & Place Order (*includes UC9, UC2*)

Field	Detail
Actor	Registered User
Precondition	Cart is non-empty; user is logged in
Main Flow	User confirms order → Order record created → Stock decremented → Cart cleared
Alternate Flow	Stock insufficient → Error shown; order not placed
Postcondition	Order persisted with status "Pending"
FR References	FR22, FR23, FR24, FR25

UC11 — View Order History

Field	Detail
Actor	Registered User
Precondition	User is logged in
Main Flow	User visits Orders page → System retrieves and lists past orders

Field	Detail
Postcondition	Order list displayed with statuses
FR References	FR26, FR28

UC12 — View Order Details (*extends UC11*)

Field	Detail
Actor	Registered User
Precondition	User is on order history page
Main Flow	User clicks an order → System shows Order ID, items, total, date, and status
Postcondition	Full order detail displayed
FR References	FR27, FR28

UC13 — Add Product (*includes UC2*)

Field	Detail
Actor	Admin User
Precondition	Admin is authenticated
Main Flow	Admin fills product form (name, description, price, image, stock) → System saves to DB
Postcondition	New product visible in listing
FR References	FR9, FR12

UC14 — Edit Product (*includes UC2*)

Field	Detail
Actor	Admin User
Precondition	Product exists; admin is authenticated
Main Flow	Admin selects product → Edits fields → System updates DB
Postcondition	Updated product details reflected immediately
FR References	FR10, FR12

UC15 — Delete Product *(includes UC2)*

Field	Detail
Actor	Admin User
Precondition	Product exists; admin is authenticated
Main Flow	Admin selects delete → Confirmation prompt → Product removed from DB
Postcondition	Product no longer visible in listing
FR References	FR11

UC16 — Manage Inventory

Field	Detail
Actor	Admin User
Precondition	Admin is authenticated
Main Flow	Admin updates stock quantity of products
Postcondition	Updated stock reflected in product availability display
FR References	FR12

5. Relationship Summary

Relationship Type	Between	Meaning
«include»	UC6 → UC2	Adding to cart always requires login
«include»	UC10 → UC9	Checkout always includes cart review
«include»	UC10 → UC2	Checkout requires authentication
«include»	UC13/14/15 → UC2	All admin actions require login
«extend»	UC5 → UC4	Detail view optionally extends browsing
«extend»	UC12 → UC11	Order detail optionally extends history view

6. Summary

The use case diagram captures all **28 functional requirements** from the SRS across **5 functional modules**. The three-tier actor model (Guest, Registered User, Admin) maps cleanly onto Django's permission framework, and the include/extend relationships reflect the application's natural navigation flow — particularly the authentication dependency embedded throughout the system.